

RudinAnalysis

†NEURON†

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Chapter 1

Overview

This blueprint records the current formal development of `RudinAnalysis`. The code follows the chapter order of Rudin's *Principles of Mathematical Analysis*, starting with ordered fields and the construction of the real numbers, then moving to metric topology and the theory of numerical sequences.

The project is intentionally elementary in style: many results are reproved directly from the chosen axioms so that the formal development mirrors the book as closely as possible.

Definition 1. The present formalization is organized into three mathematical blocks: ordered algebraic structures, basic topology of metric spaces, and convergence theory for sequences and subsequences.

Chapter 2

The Real and Complex Number Systems

2.1 Ordered sets and completeness

Definition 2. For $E \subseteq \alpha$ and $u, \ell \in \alpha$:

$$u \text{ is an upper bound of } E \iff \forall x \in E, x \leq u,$$

$$\ell \text{ is a lower bound of } E \iff \forall x \in E, \ell \leq x.$$

Accordingly,

$$E \text{ is bounded above} \iff \exists u, \forall x \in E, x \leq u,$$

$$E \text{ is bounded below} \iff \exists \ell, \forall x \in E, \ell \leq x.$$

Definition 3. For $E \subseteq \alpha$ and $s, i \in \alpha$:

$$s = \sup E \iff (\forall x \in E, x \leq s) \wedge (\forall u, (\forall x \in E, x \leq u) \rightarrow s \leq u),$$

$$i = \inf E \iff (\forall x \in E, i \leq x) \wedge (\forall \ell, (\forall x \in E, \ell \leq x) \rightarrow \ell \leq i).$$

Definition 4. The completeness predicates are:

$$\forall E \subseteq \alpha, E \neq \emptyset \wedge E \text{ bounded above} \rightarrow \exists s, s = \sup E,$$

$$\forall E \subseteq \alpha, E \neq \emptyset \wedge E \text{ bounded below} \rightarrow \exists i, i = \inf E.$$

Theorem 5. For a linear order,

$$LUB \text{ property} \iff GLB \text{ property}.$$

Proof. To pass from least upper bounds to greatest lower bounds, start with a nonempty set bounded below and consider the set of all its lower bounds. Since the original set is nonempty, this new set is bounded above by any element of the original set. Applying the least-upper-bound property gives a largest lower bound, namely the infimum. The converse is obtained by the same argument with upper and lower bounds interchanged. $\square$

Theorem 6. *Any type equipped with the project's completeness structure satisfies the greatest-lower-bound property.*

Proof. The completeness structure in the project gives the least-upper-bound property by definition. By the previous equivalence, this immediately yields the greatest-lower-bound property. $\square$

Theorem 7. *Every nonempty set that is bounded above admits a supremum.*

Proof. This is exactly the least-upper-bound part of the completeness package: once the set is known to be nonempty and bounded above, the completeness axiom produces an element satisfying the characterizing properties of a supremum. $\square$

Theorem 8. *Every nonempty set that is bounded below admits an infimum.*

Proof. By the previous theorem, the ambient order has the greatest-lower-bound property. Therefore any nonempty set bounded below has an element satisfying the defining conditions of the infimum. $\square$

2.2 Field axioms and ordered fields

Definition 9. The project introduces a custom structure of field axioms, close to Rudin's presentation, in order to rederive elementary algebraic laws from first principles.

Theorem 10. *Within any type satisfying these field axioms, one has additive and multiplicative cancellation laws.*

Proof. For additive cancellation, if $a + b = a + c$, add $-a$ to both sides and simplify using the field axioms to obtain $b = c$. For multiplicative cancellation, if $a \neq 0$ and $ab = ac$, multiply both sides by a^{-1} and use associativity together with $a^{-1}a = 1$ to conclude $b = c$. $\square$

Definition 11. An ordered field is a linear order together with the project's field axioms and the expected compatibility of order with addition and multiplication by positive elements.

Theorem 12. *In an ordered field, negatives of positive elements are negative, squares of nonzero elements are positive, and inversion reverses order on positive elements.*

Proof. If $a > 0$, then $(-a) + a = 0$ forces $-a < 0$, since $-a \geq 0$ would make $0 = (-a) + a > 0$. For a nonzero a , either $a > 0$ or $-a > 0$; in both cases a^2 is a product of positive numbers, so $a^2 > 0$. Finally, if $0 < a < b$, then multiplying by the positive number $(ab)^{-1}$ preserves order and gives $1/b < 1/a$. $\square$

2.3 Irrationality and Dedekind reals

Theorem 13. *The formal development proves that there is no rational number whose square is equal to 2.*

Proof. Assume $r = m/n$ in lowest terms and $r^2 = 2$. Then $m^2 = 2n^2$, so m^2 is even, hence m is even. Writing $m = 2k$ and substituting back gives $n^2 = 2k^2$, so n is also even. This contradicts that m/n was in lowest terms. $\square$

Definition 14. The real numbers are constructed as Dedekind cuts of rational numbers.

Theorem 15. *The ordered set of Dedekind reals satisfies the completeness axiom used elsewhere in the project.*

Proof. Given a nonempty family of Dedekind cuts bounded above, take the union of all cuts in the family. One checks that this union is again a Dedekind cut, contains every member of the family, and is the least such upper bound. Hence Dedekind reals satisfy the least-upper-bound property. $\square$

Theorem 16. *Addition on Dedekind reals is commutative and associative, and every element has an additive inverse.*

Proof. Addition is defined cutwise from rational addition. The basic algebraic laws reduce to the corresponding laws in $\mathbb{Q}$, together with verification that the resulting sets are still Dedekind cuts. The additive inverse is obtained by reflecting the cut appropriately, and then one checks that adding a cut to its inverse gives the zero cut. $\square$

Theorem 17. *The Dedekind reals carry the expected ordered-field operations: multiplication is commutative and associative, distributes over addition, and every nonzero element admits a multiplicative inverse.*

Proof. Multiplication is again defined from rational multiplication, with separate care for signs and positivity. The field laws are inherited from the rational field after checking that the cut constructions are well defined. For a nonzero cut, the inverse is built from reciprocals of rational approximations, and the defining properties show that multiplying by this inverse returns the unit cut. $\square$

Chapter 3

Basic Topology

3.1 Metric preliminaries

Definition 18. For a metric space (X, d) , $x \in X$, and $r > 0$:

$$B(x, r) = \{y \in X : d(x, y) < r\},$$

$$\overline{B}(x, r) = \{y \in X : d(x, y) \leq r\},$$

$$B^*(x, r) = \{y \in X : 0 < d(x, y) < r\}.$$

Definition 19. The basic notions are encoded as:

$$E \text{ bounded} \iff \exists x \in X \exists R > 0, E \subseteq B(x, R),$$

$$p \text{ is a limit point of } E \iff \forall r > 0, (B^*(p, r) \cap E) \neq \emptyset,$$

$$(x_n) \text{ bounded} \iff \exists x \in X \exists R > 0, \forall n, x_n \in B(x, R).$$

Theorem 20. For $E \subseteq X$,

$$E \text{ is open} \iff \forall x \in E, \exists r > 0, B(x, r) \subseteq E.$$

Proof. This is the definition of openness in a metric space. A set is open exactly when each of its points is an interior point, and that means precisely that some open ball centered at that point lies entirely inside the set. $\square$

Theorem 21. Every open ball is an open set.

Proof. Take $y \in B(x, r)$. Then $d(x, y) < r$, so the difference $r - d(x, y)$ is positive. If z lies in a small enough ball around y , the triangle inequality gives $d(x, z) < r$, so $z \in B(x, r)$. Thus every point of $B(x, r)$ contains a smaller ball still inside $B(x, r)$. $\square$

Theorem 22. Every deleted ball is open.

Proof. If $y \in B^*(x, r)$, then $0 < d(x, y) < r$. Choose a radius smaller than both $d(x, y)$ and $r - d(x, y)$. Any point z in that smaller ball around y remains within distance r of x by the triangle inequality, and also stays distinct from x by the reverse triangle estimate. Hence a whole ball around y lies in the deleted ball. $\square$

Theorem 23. *If p is a limit point of E , then*

$$\forall r > 0, (B(p, r) \cap E) \text{ is infinite.}$$

Proof. Fix $r > 0$. If $B(p, r) \cap E$ were finite, let its points other than possibly p be $x_1, \dots, x_n$. Since each $x_i \in E$, the distances $d(p, x_i)$ are positive, so we may choose $\rho > 0$ smaller than all of them and also smaller than r . Then the deleted ball $B^*(p, \rho)$ misses every x_i , contradicting that every deleted neighborhood of a limit point meets E . $\square$

3.2 Compactness

Definition 24. The project packages the finite-subcover language needed to compare Rudin's compactness with the compactness API from `mathlib`.

Theorem 25. *For $K \subseteq X$,*

$$K \text{ compact} \iff \forall \{U_i\}_{i \in I}, K \subseteq \bigcup_{i \in I} U_i \wedge (\forall i, U_i \text{ open}) \rightarrow \exists i_1, \dots, i_n, K \subseteq U_{i_1} \cup \dots \cup U_{i_n}.$$

Proof. This is the standard finite-subcover characterization of compactness. One direction says that compactness precisely means every open cover can be reduced to finitely many members; the other direction simply repackages that same condition in the language used by the project. $\square$

Theorem 26. *Compactness behaves well under induced topologies, and compact subsets of metric spaces are closed.*

Proof. Passing to a subspace does not change what it means for an open cover of the subset to have a finite subcover, so compactness is preserved by the induced topology. For closedness in a metric space, take a point outside a compact set. Distances from that point to elements of the compact set attain a positive minimum, which yields an open ball disjoint from the compact set. Thus the complement is open. $\square$

Theorem 27. *For a compact set, any family of closed subsets with the finite intersection property has nonempty total intersection.*

Proof. Assume the total intersection were empty. Then the complements of the closed sets form an open cover of the compact set. By compactness, finitely many of these complements already cover the compact set, which means the corresponding finite family of closed sets has empty intersection. This contradicts the finite intersection property. $\square$

Theorem 28. *Every infinite subset of a compact metric space has a limit point in that compact set.*

Proof. Suppose an infinite subset E had no limit point in the compact set K . Then each point of K would have a neighborhood meeting E in at most finitely many points. These neighborhoods form an open cover of K , so compactness gives a finite subcover. But then E , being a union of finitely many finite sets, would itself be finite, a contradiction. $\square$

Chapter 4

Numerical Sequences and Series

4.1 Convergent sequences

Definition 29. For a sequence (x_n) in a metric space and $p \in X$,

$$x_n \rightarrow p \iff \forall \varepsilon > 0, \exists N, \forall n \geq N, d(x_n, p) < \varepsilon.$$

The blueprint also uses the bundled predicates “ (x_n) is convergent” and “ $\lim x_n$ ”.

Theorem 30. For a sequence (x_n) and $p \in X$,

$$x_n \rightarrow p \iff \forall r > 0, \exists N, \forall n \geq N, x_n \in B(p, r).$$

Proof. This is just a reformulation of the definition of convergence: the inequality $d(x_n, p) < r$ says exactly that x_n belongs to the open ball $B(p, r)$. $\square$

Theorem 31. Limits of sequences in metric spaces are unique.

Proof. Assume $x_n \rightarrow p$ and $x_n \rightarrow q$. If $p \neq q$, let $\varepsilon = \frac{1}{2}d(p, q) > 0$. Eventually the sequence lies in both $B(p, \varepsilon)$ and $B(q, \varepsilon)$. For such an index n , the triangle inequality gives $d(p, q) \leq d(p, x_n) + d(x_n, q) < \varepsilon + \varepsilon = d(p, q)$, a contradiction. $\square$

Theorem 32.

$$x_n \rightarrow p \implies (x_n) \text{ is bounded.}$$

Proof. Choose N so that $n \geq N$ implies $x_n \in B(p, 1)$. The finitely many initial terms $x_0, \dots, x_{N-1}$ are bounded as well, so taking a radius large enough to contain them together with $B(p, 1)$ shows that the whole sequence is bounded. $\square$

Theorem 33. If p is a limit point of E , then there exists a sequence (x_n) such that

$$\forall n, x_n \in E, \quad x_n \neq p, \quad x_n \rightarrow p.$$

Proof. Because p is a limit point, each deleted ball $B^*(p, 1/(n+1))$ meets E . Choose $x_n \in E \cap B^*(p, 1/(n+1))$. Then $x_n \neq p$ and $d(x_n, p) < 1/(n+1)$, which tends to 0. Hence $x_n \rightarrow p$. $\square$

Theorem 34. For complex-valued sequences,

$$x_n \rightarrow x, y_n \rightarrow y \implies x_n + y_n \rightarrow x + y, \quad x_n y_n \rightarrow xy,$$

$$x_n \rightarrow x, c \in \mathbb{C} \implies cx_n \rightarrow cx, \quad x_n + c \rightarrow x + c.$$

Proof. These are the usual ε - N estimates. For sums and scalar translates, one splits the error using the triangle inequality. For products, write $x_n y_n - xy = x_n(y_n - y) + y(x_n - x)$ or an equivalent decomposition; convergence gives control of each term, and boundedness of convergent sequences keeps the remaining factors under control. $\square$

4.2 Subsequences and compactness

Definition 35. A subsequence of (x_n) is given by a strictly increasing map

$$\phi : \mathbb{N} \rightarrow \mathbb{N}, \quad \phi(m) < \phi(n) \text{ whenever } m < n,$$

and the induced sequence is $(x_{\phi(n)})$.

Theorem 36. If $x_n \rightarrow p$ and ϕ is strictly increasing, then

$$x_{\phi(n)} \rightarrow p.$$

Proof. Given $\varepsilon > 0$, choose N such that $n \geq N$ implies $d(x_n, p) < \varepsilon$. Since ϕ is strictly increasing, we have $\phi(n) \geq n$, so for all $n \geq N$ also $\phi(n) \geq N$. Hence $d(x_{\phi(n)}, p) < \varepsilon$ eventually. $\square$

Theorem 37. If (x_n) is a sequence in a compact metric space, then there exist a strictly increasing $\phi : \mathbb{N} \rightarrow \mathbb{N}$ and p such that

$$x_{\phi(n)} \rightarrow p.$$

In particular, every bounded sequence in $\mathbb{R}^m$ has a convergent subsequence.

Proof. If the sequence takes only finitely many values, one value occurs infinitely often, and the corresponding constant subsequence converges. Otherwise its range is an infinite subset of a compact space, so by Bolzano–Weierstrass it has a limit point p . The previous theorem then produces a sequence of distinct range points converging to p , and choosing indices where those values occur yields a convergent subsequence of the original sequence. In $\mathbb{R}^m$, a bounded sequence has compact closure, so the same argument applies. $\square$

Definition 38. The set of subsequential limits of (x_n) is

$$\text{SubLim}(x_n) = \{p \in X : \exists \phi : \mathbb{N} \rightarrow \mathbb{N} \text{ strictly increasing, } x_{\phi(n)} \rightarrow p\}.$$

Theorem 39. The set of subsequential limits of a sequence is closed.

Proof. Let p_k be subsequential limits with $p_k \neq p$. For each k , choose a subsequence converging to p_k . A diagonal selection picks one term far enough along the k -th subsequence so that it is close both to p_k and hence to p . The resulting extracted subsequence converges to p , so p is itself a subsequential limit. $\square$

4.3 Cauchy sequences and completeness

Definition 40. For a sequence (x_n) in a metric space:

$$(x_n) \text{ is Cauchy} \iff \forall \varepsilon > 0, \exists N, \forall m, n \geq N, d(x_m, x_n) < \varepsilon.$$

Its N -th tail set is

$$T_N = \{x_n : n \geq N\}.$$

Completeness is expressed by

$$X \text{ complete} \iff \text{every Cauchy sequence in } X \text{ converges.}$$

Theorem 41.

$$(x_n) \text{ is Cauchy} \iff \text{diam}(T_N) \rightarrow 0.$$

Moreover,

$$\text{diam}(\overline{E}) = \text{diam}(E).$$

Proof. The sequence is Cauchy exactly when, for every $\varepsilon > 0$, all sufficiently far tail terms lie within ε of one another; this is precisely the statement that the diameters of the tail sets tend to 0. For the closure statement, adding limit points does not enlarge any upper bound for pairwise distances, so the diameter of a set and of its closure are the same. $\square$

Theorem 42. If (K_n) is a decreasing sequence of nonempty compact sets with

$$K_{n+1} \subseteq K_n \quad \text{and} \quad \text{diam}(K_n) \rightarrow 0,$$

then

$$\bigcap_{n=0}^{\infty} K_n = \{p\}$$

for some point p .

Proof. The nesting implies every finite intersection is nonempty, so compactness and the finite intersection property give a point p in the total intersection. If q is another point in the intersection, then both p and q lie in every K_n , hence $d(p, q) \leq \text{diam}(K_n)$ for all n . Since these diameters tend to 0, we get $d(p, q) = 0$, so $p = q$. $\square$

Theorem 43.

$$x_n \rightarrow p \implies (x_n) \text{ is Cauchy.}$$

Also,

$$X \text{ compact} \implies \text{every Cauchy sequence in } X \text{ converges,}$$

and in particular every Cauchy sequence in $\mathbb{R}^m$ converges.

Proof. If $x_n \rightarrow p$, then eventually all terms lie in a ball of radius $\varepsilon/2$ around p , so any two sufficiently late terms are within ε of each other; hence the sequence is Cauchy. Conversely, in a compact space every sequence has a convergent subsequence. A Cauchy sequence and any convergent subsequence must have the same limit for the whole tail, so the full sequence converges to that subsequential limit. Euclidean space inherits this once bounded closed sets are compact in the formalization being used. $\square$

Theorem 44.

$$X \text{ compact} \implies X \text{ complete,}$$

$$\mathbb{R}^m \text{ is complete,}$$

$$X \text{ complete and } E \subseteq X \text{ closed} \implies E \text{ complete.}$$

Proof. The first statement is just the preceding theorem: in a compact space every Cauchy sequence converges. The Euclidean case is a special instance. For a closed subset E of a complete space X , any Cauchy sequence in E converges in X to some point p , and closedness of E forces $p \in E$. Thus the sequence also converges in E . $\square$

4.4 Monotone sequences

Definition 45. For a real sequence (s_n) :

$$\text{increasing} \iff \forall m \leq n, s_m \leq s_n,$$

$$\text{decreasing} \iff \forall m \leq n, s_n \leq s_m,$$

$$\text{monotone} \iff \text{increasing or decreasing}.$$

Theorem 46. For a monotone real sequence (s_n) ,

$$(s_n) \text{ converges} \iff (s_n) \text{ is bounded}.$$

Proof. Every convergent sequence is bounded, so only the converse needs proof. If (s_n) is increasing and bounded above, let $s = \sup\{s_n : n \in \mathbb{N}\}$. Then for every $\varepsilon > 0$, $s - \varepsilon$ cannot be an upper bound, so some term satisfies $s - \varepsilon < s_N \leq s$. Monotonicity gives $s - \varepsilon < s_n \leq s$ for all $n \geq N$, hence s_n os. The decreasing case is analogous, using the infimum. $\square$